



The Lotus 88 dominated Formula 1 by leveraging aerodynamics to produce tremendous grip

Because drag increases with the square of vehicle speed, aerodynamic efficiency becomes one of the most important factors dictating race car performance. Although the concept of inverted wing was first employed in the IndyCar arena, it was Colin Chapman who turned aerodynamics into a gamebreaker with the Lotus 88 Formula 1 car.

The car was the first one to employ the unique twin-chassis system comprising of an independently sprung inner chassis housing the cockpit and the outer one comprising the aero package. The idea was to make the bottom absolutely flat to speed up air flow and bolstered with skirting to enhance the effect. The outer chassis was equipped with wings designed to slow down airflow. The resulting difference in airflow employs the Bernoulli effect to create a tremendous amount of downforce. Enough of it to give the car an unfair advantage. The ensuing sweeping victories initially had everyone puzzled, but the FIA eventually caught up to Chapman's genius and banned the car from competing. To this day, FIA lays down strict regulation regarding front diffuser/rear wing areas and the overall downforce generated.

Aerodynamics might be a major aspect, but you can't really go fast without absolute power. Formula 1 engineers are used to tweaking turbochargers to maximum power for qualifying, which involves far fewer laps as compared to the main race. This is mainly because engine endurance isn't a major consideration given the circumstances. BMW, however, managed to squeeze out 1500 bhp from a puny 1.5 litre engine way back in the '80s. In comparison, a 2016 Mercedes Formula 1 car can only muster 900 bhp. Interestingly, BMW wasn't exactly

employing rocket science to beat everyone at the power game. The engineers only chose to do away with durability in order to push the engine to the very precipices of performance. The engine would be allowed to run hot enough to last long enough to complete qualifying, after which it would be summarily discarded. FIA soon figured out what BMW was doing and enforced stricter rules on engine changes.

SPY GADGETS AND EXAMS

Passing the act of cheating off as human ingenuity may seem fairly kosher, but applying the same logic to exams is a slippery slope few would like to go down. While the former may entail the triumph of the resourceful rule breakers at its worst, the latter involves incompetent cheaters getting an undeserved chance to toy with your life and limb. Be it medical or engineering exams, cheaters employ a variety of high-tech telecommunication gadgets to game the system. A sophisticated cheating racket in Thailand involves professional tutors equipping candidates with specialised glasses containing hidden earphones and tiny micro-cameras; all of which are linked to smartwatches that allow the tutors to livestream exam solutions back to the candidates. Other slightly less sophisticated setups involve spy pen cameras used to relay questions to tutors, who then communicate the answers through trusty Nokia feature phones paired with discreet earphones.



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CHEATING DEMOCRACY

Just when you think cheating can't get any worse, we are faced with the sort of scandal that could potentially lead to the very failure of democracy. This is the sort of miscarriage of justice that eventually leads to the fall of civilisa-

tions. In more specific terms, we refer to Electronic Voting Machine (EVM) fraud. While paper voting is the most secure means to conduct polling, it's quite wasteful, slow, and tedious. Unfortunately, there's no clear security standard enforced within the myriad EVM standards that cater to nearly 70-percent of the US electorate.

EVM system programmers and hackers alike have testified in courts about the susceptibility of these machines to tampering either before or after the voting process. In fact, most of these EVM don't even use any form of encryption and are vulnerable to vote manipulation as well as multiple vote fraud. And if you think the ability to thwart and bypass these supposedly secure EVMs would require expensive and sophisticated equipment, think again because hackers at Symantec Security Response have already demonstrated that these machines can be hacked by devices that are easy to acquire and cost no more than \$15 a pop.



Rigged EVMs: an illusion of democracy

GEEKS SHALL INHERIT THE EARTH

It is rightly said that geeks shall inherit the Earth. Or was it the meek? At any rate, it is not just our thirst for knowledge and excellence that has made us the dominant species. Our ability to think laterally to solve difficult problems has helped us evolve against deep odds. The tendency to cheat is a direct byproduct of this innate instinct. It is amply evident that its ramifications change depending on how we go about it. Cheating becomes ingenuity and innovation when applied to motorsport. But at the same time, it's a grave social and political evil when education and democracy is involved. At the end of the day, the power to choose between what's right and what's convenient squarely rests upon the shoulders of the same enterprising geeks. 